

Clustering-Guided Contrastive Learning for Clinically Aware Chest X-ray Image–Report Retrieval

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Abstract

Cross-modal contrastive learning for chest X-ray image-report retrieval commonly suffers from the *false-negative* problem: image-report pairs belonging to different patients but sharing the same pathological findings are treated as hard negatives, introducing incorrect gradient pressure and degrading clinically meaningful retrieval performance. We propose a **clustering-guided contrastive learning framework** to mitigate this issue on the IU-Xray dataset. Our model combines a **SwinV2-Base** image encoder with a **Bio_ClinicalBERT** text encoder, projecting both modalities into a shared 512-dimensional embedding space through deep MLP projection heads. Multi-view study-level representations are formed via *attention pooling* over frontal and lateral views. Training follows a three-phase curriculum: (i) strict image-report alignment using multi-positive InfoNCE, (ii) soft-target blending with IDF-weighted Jaccard pathology similarity, and (iii) a clinical supervised contrastive loss with a gradual curriculum schedule that pulls same-disease, non-normal pairs together without compromising exact-pair retrieval. On a fixed patient-stratified split of IU-Xray (377 test studies), our model achieves a **clinical-valid mean R@1 of 59.86%**, cluster mean R@1 of 69.36%, and strict mean R@1 of 3.85% (MRR 9.01%) on the test set. Qualitative analysis identifies **157 clinical rescue** cases in top-1 (64 image-to-text + 93 text-to-image), where the model retrieves a clinically related result from a different patient, directly validating the false-negative mitigation objective.

Keywords: medical image-text retrieval, contrastive learning, false negatives, pathology clustering, chest X-ray, SwinV2, ClinicalBERT, curriculum learning.

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1. Introduction

Automated medical image–text retrieval aims to associate diagnostic images with their corresponding clinical reports in a shared embedding space. Practical applications include case-based reasoning support, automated report generation, and clinical decision support systems (Jing et al., 2018).

Multimodal contrastive learning in the style of CLIP (Radford et al., 2021) has emerged as the dominant paradigm for this task. Methods such as ConVIRT (Zhang et al., 2022) and GLORIA (Huang et al., 2021) train a model to bring together *same-patient* image–report pairs (positives) while pushing apart *different-patient* pairs (negatives).

The False-Negative Problem. In clinical practice, patients from different studies frequently present identical or highly similar pathological findings. As illustrated in Figure 1, standard InfoNCE treats these *clinically related* but *patient-different* pairs as hard negatives, penalizing the model for placing them nearby. This produces incorrect gradient signals and suppresses the model’s ability to perform semantically meaningful, disease-level retrieval—the very capability that is most valuable in a clinical context.

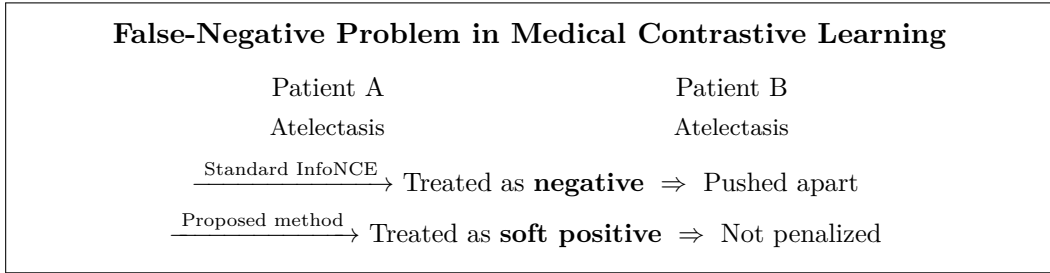


Figure 1: The false-negative problem in medical contrastive learning. Two patients sharing the same pathology (Atelectasis) are incorrectly penalized by standard InfoNCE. Our method assigns a soft positive target proportional to their pathological similarity.

Contributions. This paper makes the following contributions:

- (i) A **study-level multi-view retrieval pipeline** for IU-Xray using attention pooling over frontal and lateral X-ray views.
- (ii) A **hierarchical clustering loss** based on IDF-weighted Jaccard pathology similarity, generating soft training targets that reduce false-negative pressure from same-disease different-patient pairs.
- (iii) A **clinical supervised contrastive loss** with a curriculum schedule that gradually introduces and then decays disease-cluster supervision, preserving exact-pair retrieval ability.
- (iv) A **three-tier evaluation framework**—strict, cluster, and clinical-valid R@K—where *clinical-valid* R@K is the primary metric directly measuring false-negative mitigation.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3

describes the proposed method. Section 4 presents experimental settings. Section 5 reports results. Section 6 discusses findings and limitations. Section 7 concludes.

2. Related Work

2.1. Multimodal Contrastive Learning

CLIP (Radford et al., 2021) established the foundation for vision–language pretraining by optimizing InfoNCE (Oord et al., 2018) over 400 million web-crawled image–caption pairs. ConVIRT (Zhang et al., 2022) pioneered the application of contrastive learning to radiology, aligning chest X-rays with clinical reports at the global level. GLoRIA (Huang et al., 2021) extended this with local attention-weighted sub-regional alignment. BiomedCLIP (Zhang et al., 2023) scaled medical vision–language pretraining to 15 million figure–caption pairs from PubMed Central, achieving strong zero-shot performance across multiple biomedical tasks.

2.2. False Negatives in Contrastive Learning

The false-negative problem has been studied in the general contrastive learning literature. Khosla et al. (2020) introduced Supervised Contrastive Learning (SupCon), which uses class labels to identify multiple valid positives per query in a batch. Chuang et al. (2020) proposed a debiased estimator that corrects for the probability of sampling true negatives. Robinson et al. (2021) studied hard negative sampling while explicitly controlling false-negative contamination.

In radiology specifically, false negatives are particularly severe because: (1) many patients share common pathologies (*e.g.* cardiomegaly, atelectasis), (2) the dataset is heavily label-imbalanced (IU-Xray: 57.88% no-finding), and (3) pathological semantics span both visual appearance and clinical language simultaneously. To our knowledge, no prior work has proposed an IDF-weighted Jaccard soft target combined with a clinical curriculum for this problem on IU-Xray.

2.3. Medical Visual–Linguistic Encoders

SwinV2 (Liu et al., 2022) extends Swin Transformer with post-normalization layer scaling and a continuous relative position bias, enabling resolution transfer from low to high resolution (192→384) without degradation. Bio_ClinicalBERT (Alsentzer et al., 2019) is a BERT model further pretrained on MIMIC-III clinical notes, making it well-suited for the register of radiology reports.

2.4. Chest X-ray Image–Report Retrieval

The IU-Xray dataset (Demner-Fushman et al., 2016) is the most widely used benchmark for radiology report generation and retrieval. Prior retrieval studies on IU-Xray typically report only strict R@K (same-patient ground truth), which does not capture whether the model can retrieve clinically related results from different patients. This paper introduces *clinical-valid* R@K to address this gap and provide a more complete picture of retrieval quality.

3. Proposed Method

Figure 2 provides an overview of the model architecture and training pipeline.

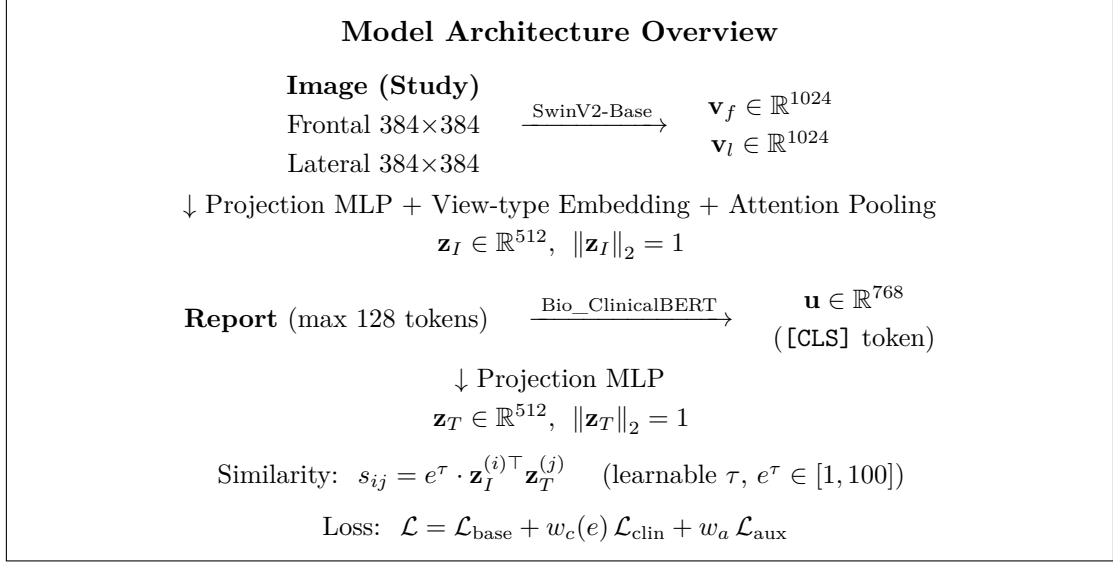


Figure 2: Architecture overview. Both image and text modalities are projected into a shared 512-d ℓ_2 -normalized embedding space. The loss combines a base contrastive term, a clinical supervised contrastive term with curriculum weight $w_c(e)$, and an auxiliary pathology prediction term.

3.1. Data Representation

Each sample in IU-Xray corresponds to one patient *study*, comprising:

- **Images:** up to one frontal and one lateral X-ray, stored as 384 × 384 RGB PNG after black-padding to preserve the full field of view without geometric distortion.
- **Report:** the concatenation of the **findings** and **impression** fields after text normalization (lowercasing, removal of placeholders, section headers, and short texts below 10 words).
- **Pathology vector:** $\mathbf{d} \in \{0, 1\}^{12}$, derived from 14 CheXpert labels (Irvin et al., 2019) via negation-aware keyword matching. “No Finding” is excluded; “Enlarged Cardiomediastinum” is merged into “Cardiomegaly” by taking the maximum.

A study is considered **normal** when $\sum_k d_k = 0$.

3.2. Model Architecture

3.2.1. Image Encoder

We use **SwinV2-Base** (Liu et al., 2022), specifically the checkpoint `microsoft/swinv2-base-patch4-window` pretrained on ImageNet-22K and fine-tuned at 384×384 resolution. The `pooler_output` with dimension $\mathbb{R}^{1024}$ is used. Gradient checkpointing is enabled to reduce GPU memory consumption.

3.2.2. Text Encoder

We use **Bio_ClinicalBERT** (Alsentzer et al., 2019) (emilyalsentzer/Bio_ClinicalBERT), pretrained on MIMIC-III clinical notes. Reports are truncated to 128 tokens; the [CLS] embedding from the final hidden layer ($\mathbb{R}^{768}$) is used as the report representation.

3.2.3. Projection Head

Both modalities pass through an identical two-layer MLP:

$$\mathbf{z} = \frac{\text{MLP}(\mathbf{h})}{\|\text{MLP}(\mathbf{h})\|_2}, \quad (1)$$

where

$$\text{MLP}(\mathbf{h}) = \mathbf{W}_2 \text{Dropout}_{0.2}(\text{GELU}(\text{BN}(\mathbf{W}_1 \mathbf{h} + \mathbf{b}_1))) + \mathbf{b}_2, \quad (2)$$

with $\mathbf{W}_1 \in \mathbb{R}^{1024 \times d_{\text{in}}}$, $\mathbf{W}_2 \in \mathbb{R}^{512 \times 1024}$, BN = Batch Normalization. The output $\mathbf{z} \in \mathbb{R}^{512}$ is ℓ_2 -normalized.

Similarity between an image–text pair is computed as:

$$s_{ij} = e^\tau \cdot \mathbf{z}_I^{(i)\top} \mathbf{z}_T^{(j)}, \quad (3)$$

where τ is a learnable scalar (*logit_scale*) clamped so that $e^\tau \in [1, 100]$, following Radford et al. (2021).

3.2.4. Multi-View Attention Pooling

Each view (frontal / lateral) is independently encoded by SwinV2 and the projection head to obtain $\mathbf{z}_k \in \mathbb{R}^{512}$. A *view-type embedding* $\mathbf{e}_k^{\text{type}} \in \mathbb{R}^{512}$ (learned, one per view type: frontal, lateral, other, pad) is added. The study embedding is computed by:

$$\alpha_k = \frac{\exp(a(\mathbf{z}_k + \mathbf{e}_k^{\text{type}}))}{\sum_{k'} \exp(a(\mathbf{z}_{k'} + \mathbf{e}_{k'}^{\text{type}}))}, \quad \mathbf{z}_I = \sum_{k=1}^K \alpha_k \mathbf{z}_k, \quad (4)$$

where $a(\cdot) = \mathbf{w}_2^\top \text{GELU}(\mathbf{W}_1 \cdot)$ is a lightweight attention network ($\mathbf{W}_1 \in \mathbb{R}^{256 \times 512}$, $\mathbf{w}_2 \in \mathbb{R}^{256}$), and $K \leq 2$ is the number of valid views (indicated by *view_mask*). Missing views are zero-padded and masked out. This design allows the model to learn view importance adaptively rather than relying on a fixed average.

3.2.5. Auxiliary Pathology Heads

Four lightweight MLP heads branch off the projected embeddings: *img_pathology_head* and *img_normal_head* (from $\mathbf{z}_I$) predict 12 pathology labels and a normal/abnormal binary label respectively; *txt_pathology_head* and *txt_normal_head* do the same from $\mathbf{z}_T$. These heads enforce pathology-informative embeddings at a small auxiliary weight ($w_a = 0.02$), ensuring they do not dominate the retrieval objective.

3.3. Hierarchical Clustering Loss

3.3.1. Phase 1 — Multi-Positive InfoNCE (Epochs 1–11)

Before the clustering phase begins, the base loss is multi-positive InfoNCE:

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(s_{ii})}{\sum_{j=1}^N \exp(s_{ij})}, \quad (5)$$

computed bidirectionally (image→text and text→image). Since s_{ij} in (3) already incorporates the learned temperature, no additional temperature division is applied. The ground-truth positive for query i is the entry sharing the same `patient_id`.

3.3.2. Phase 2 — Hierarchical Clustering Loss (Epochs 12+)

From epoch 12 onward, the hard binary ground-truth is replaced by a *soft target* that blends exact-patient matching with pathological similarity:

$$t_{ij} = (1 - \alpha) \mathbb{I}[p_i = p_j] + \alpha S_{ij}^{\text{IDF}}, \quad \alpha = 0.35. \quad (6)$$

The **IDF-weighted Jaccard similarity** between disease vectors $\mathbf{d}_i, \mathbf{d}_j \in \{0, 1\}^{12}$ is defined as:

$$S^{\text{IDF}}(\mathbf{d}_i, \mathbf{d}_j) = \frac{\sum_{k=1}^{12} w_k \cdot \min(d_{ik}, d_{jk})}{\sum_{k=1}^{12} w_k \cdot \max(d_{ik}, d_{jk})}, \quad (7)$$

where the IDF weight for disease k is:

$$w_k = \frac{\log(1/f_k)}{\sum_{k'} \log(1/f_{k'})}, \quad (8)$$

and f_k is the prevalence of disease k in the training set. This weighting assigns higher importance to rare conditions (*e.g.* Pneumothorax $f=2.03\%$, Fracture $f=1.82\%$) over more common ones (*e.g.* Cardiomegaly $f=9.00\%$), reflecting the greater clinical discriminative value of rare findings.

Healthy cluster bonus. When both samples have $\sum_k d_k^{(i)} = \sum_k d_k^{(j)} = 0$ (both normal), the Jaccard similarity is undefined. We add a fixed similarity bonus:

$$S_{ij}^{\text{IDF}} \leftarrow 0.4 \cdot \mathbb{I}[\text{normal}(i)] \cdot \mathbb{I}[\text{normal}(j)], \quad (9)$$

preventing the majority normal class (57.88% of data) from scattering in embedding space.

3.3.3. Phase 3 — Clinical Supervised Contrastive Loss (Epochs 15+)

This is the core contribution targeting false-negative mitigation.

Clinical mask. For each pair (i, j) , the clinical positive indicator is:

$$M_{ij}^{\text{clin}} = \underbrace{\mathbb{I}[\mathbf{d}_i \cdot \mathbf{d}_j > 0]}_{\text{disease overlap}} \cdot \mathbb{I}[i \neq j] \cdot \underbrace{\mathbb{I}[\text{non-normal}(i)] \cdot \mathbb{I}[\text{non-normal}(j)]}_{\text{both abnormal}}. \quad (10)$$

This selects pairs from *different* studies that share at least one disease label and are both abnormal. Normal-normal pairs are excluded to avoid trivially inflating the metric.

Loss. For each non-normal query i with at least one clinical positive ($P_i = \{j : M_{ij}^{\text{clin}} = 1\}$, $|P_i| > 0$):

$$\mathcal{L}_{\text{clin}} = -\frac{1}{|\mathcal{Q}|} \sum_{i \in \mathcal{Q}} \frac{1}{|P_i|} \sum_{j \in P_i} \log \frac{\exp(s_{ij})}{\sum_{k=1}^N \exp(s_{ik})}, \quad (11)$$

where $\mathcal{Q} = \{i : |P_i| > 0\}$. The loss is computed bidirectionally.

3.3.4. Auxiliary Pathology Loss

$$\mathcal{L}_{\text{aux}} = \frac{1}{4} [\text{BCE}(\hat{\mathbf{d}}_I, \mathbf{d}) + \text{BCE}(\hat{n}_I, n) + \text{BCE}(\hat{\mathbf{d}}_T, \mathbf{d}) + \text{BCE}(\hat{n}_T, n)] + 0.25 [\text{MSE}(\sigma(\hat{\mathbf{d}}_I), \sigma(\hat{\mathbf{d}}_T)) + \text{MSE}(\sigma(\hat{n}_I), \sigma(\hat{n}_T))], \quad (12)$$

where σ is sigmoid. The MSE term enforces cross-modal consistency in pathology predictions.

Total loss:

$$\mathcal{L} = \mathcal{L}_{\text{base}} + w_c(e) \mathcal{L}_{\text{clin}} + w_a \mathcal{L}_{\text{aux}}, \quad w_a = 0.02. \quad (13)$$

3.4. Clinical Curriculum Schedule

Activating $\mathcal{L}_{\text{clin}}$ at full weight from the start risks pulling the model toward broad disease clusters before it has learned exact-pair alignment. We therefore apply a ramp-up / hold / decay curriculum:

$$w_c(e) = \begin{cases} 0 & e < 15, \\ 0.03 + 0.09 \frac{e-15}{10} & 15 \leq e \leq 25, \\ 0.12 & 26 \leq e \leq 60, \\ 0.08 & 61 \leq e \leq 105, \\ 0.04 & 106 \leq e \leq 150. \end{cases} \quad (14)$$

Table 1 summarizes the purpose of each phase.

Table 1: Clinical curriculum schedule and training phase objectives.

Epochs	Primary objective	w_c	Phase
1–5	Projection heads only (encoders frozen)	0.00	Warm-up
6–11	Multi-positive InfoNCE, all unfrozen	0.00	Strict align
12–14	Soft clustering targets activated	0.00	Cluster init
15–25	Clinical loss ramping	0.03→0.12	Clinical ramp
26–60	Peak clinical supervision	0.12	Clinical peak
61–105	Clinical weight decay	0.08	Decay
106–150	Final strict recovery	0.04	Fine-tune

The rationale is validated by our results (Section 5): strict R@1 reaches its validation peak at epoch 120, *after* the clinical decay begins at epoch 60, confirming that reducing w_c late in training allows the model to recover precise image–report alignment.

4. Experiments

4.1. Dataset

The **IU-Xray** dataset (Demner-Fushman et al., 2016) (Indiana University Chest X-ray Collection) contains chest radiographs collected from Indiana University hospital network, paired with de-identified radiology reports.

Preprocessing. Images are converted to grayscale, then to RGB, padded to a square aspect ratio with a black background, and saved at 384×384 pixels to match SwinV2-Base’s native resolution without cropping. Reports are lower-cased, stripped of section headers (*e.g.* “findings:”, “impression:”) and age/gender tokens, and filtered to retain only reports with ≥ 10 words. Pathology labels are generated by negation-aware keyword matching over 14 CheXpert categories and then reduced to a 12-dimensional disease vector.

Dataset statistics are summarized in Table 2.

Table 2: IU-Xray dataset statistics after preprocessing.

Statistic	Value
Total images (384×384 PNG)	7,322
Unique patient / study IDs	3,772
Unique reports	3,018
Words per report (min / mean / max)	10 / 37.27 / 223
Frontal views	3,738
Lateral views	3,584
<i>Patient-stratified split (seed = 42)</i>	
Train (study-level)	3,018
Validation	377
Test	377

Pathology label prevalence is shown in Table 3.

Table 3: Prevalence of 14 CheXpert pathology labels in IU-Xray (7,322 images).

Label	Count	%
No Finding	4,238	57.88
Cardiomegaly	659	9.00
Lung Lesion	1,014	13.85
Lung Opacity	942	12.87
Atelectasis	732	10.00
Consolidation	402	5.49
Pleural Effusion	415	5.67
Support Devices	224	3.06
Edema	185	2.53
Pneumonia	164	2.24
Pneumothorax	149	2.03
Fracture	133	1.82
Pleural Other	65	0.89
Enlarged Cardiomedastinum	17	0.23

4.2. Implementation Details

Hardware. NVIDIA A100 (40 GB VRAM), PyTorch with `torch.amp.autocast + GradScaler` mixed-precision training.

Hyperparameters are listed in Table 4.

Table 4: Training hyperparameters for the v8 run (v8_curriculum_a100_150ep_seed42_bs32_v1).

Hyperparameter	Value
Epochs	150
Batch size	32
Gradient accumulation steps	4
Effective batch size	128
Backbone freeze epochs	5
Evaluation frequency	every 5 epochs
Random seed	42
LR — image encoder	1×10^{-5}
LR — text encoder	1×10^{-4}
LR — projection heads	2×10^{-4}
Weight decay	0.01
Max gradient norm	1.0
Optimizer	AdamW
LR scheduler (post-unfreeze)	Linear warmup (5%) + CosineAnnealing
$\eta_{\min}$	10^{-8}
α (soft-target blend)	0.35
Cluster temperature	1.0
w_a (auxiliary weight)	0.02
Embedding dimension	512
Projection hidden dimension	1,024
Projection dropout	0.2

Checkpoint strategy. Three checkpoints are maintained: `best.pt` (best validation strict R@1), `best_balanced.pt` (best strict + 0.1×clinical-valid, when strict $\geq 4.0\%$), and `best_clinical_valid.pt` (best clinical-valid, when strict $\geq 3.5\%$). The primary reported checkpoint is `best.pt` (epoch 120).

Disabled components. For transparency, we note that the codebase contains *DualPrototypeBank*, *IntraClusterRankingLoss*, and *Hard Negative Mining* components, but all are *disabled* in the v8 final run (`proto_start=999`, `rank_start=999`, `mine_start=999`). The results reported here do *not* rely on these components; they are reserved for future ablation studies.

4.3. Evaluation Metrics

All evaluations are at study-level: each patient is one query, and the query embedding is the attention-pooled image embedding. We report metrics in both directions: image-to-text (i2t) and text-to-image (t2i).

M1 — Strict Retrieval. $G_{ij}^{\text{strict}} = \mathbb{I}[p_i = p_j]$. Only the same-patient study is a ground-truth hit. Metrics: R@1, R@5, R@10, MRR.

M2 — Cluster Retrieval. $G_{ij}^{\text{clus}} = \mathbb{I}[\mathbf{d}_i \cdot \mathbf{d}_j > 0] + \mathbb{I}[\text{normal}(i) \wedge \text{normal}(j)]$. Both disease-overlapping pairs and normal-normal pairs are hits.

M3 — Clinical-Valid Retrieval (primary metric). $G_{ij}^{\text{clin}} = \mathbb{I}[\mathbf{d}_i \mathbf{d}_j > 0] \wedge \mathbb{I}[\text{non-normal}(i)] \wedge \mathbb{I}[\text{non-normal}(j)]$. Computed only for non-normal queries with ≥ 1 clinical positive in the test set ($n=147$ per direction). This metric directly measures false-negative mitigation.

M4 — Clinical-All Retrieval. Same mask as M3 but evaluated over all 377 queries (including those with no clinical positive). Used as a secondary metric.

Qualitative bucket analysis. Each top-1 result is classified into one of four mutually exclusive buckets: *strict hit* (same patient), *clinical rescue* (different patient, clinical mask satisfied), *cluster rescue* (cluster mask satisfied but not clinical), or *miss*.

5. Results

5.1. Quantitative Results on the Test Set

Table 5 reports the full test-set metrics for the primary checkpoint `best.pt` (epoch 120, 377 test studies).

Table 5: Test-set retrieval performance of the proposed model (v8, `best.pt`, epoch 120). **Mean** = average of i2t and t2i. The primary metric is *clinical-valid* (shaded).

	Strict			Cluster			Clinical-Valid		
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10
i2t	3.45	10.08	17.77	68.17	79.58	85.15	49.66	60.54	69.39
t2i	4.24	12.20	18.83	70.56	97.08	99.20	70.07	95.24	97.96
Mean	3.85	11.14	18.30	69.36	88.33	92.18	59.86	77.89	83.67

Strict MRR = 9.01%; Clinical-all mean R@1 = 23.34%; Clinical-valid: 147/377 queries per direction have ≥ 1 positive.

Best validation metrics (across all epochs): strict mean R@1 = **4.91%** (epoch 120); clinical-valid mean R@1 = **52.11%** (epoch 90); cluster mean R@1 = **65.25%** (epoch 40).

5.2. Comparison with Baseline

Table 6 compares the proposed model against v7, which uses the same architecture and split but a less refined curriculum schedule.

Table 6: Comparison of v8 (proposed) vs. v7 (baseline) on the test set. Both models use the same patient-stratified split (seed=42).

Model	Strict Mean R@1	Cluster Mean R@1	Clin.-Valid Mean R@1	N
V7 (best-balanced ckpt)	3.32%	67.37%	59.86%	9
V8 (proposed, ep. 120)	3.85%	69.36%	59.86%	9
Δ	+0.53 pp	+1.99 pp	0.00 pp	

V8 improves strict mean R@1 by **+0.53 pp** and cluster mean R@1 by **+1.99 pp** while maintaining clinical-valid mean R@1 at the same high level of 59.86%. This demonstrates that the curriculum decay in v8 successfully preserves and improves exact retrieval without sacrificing clinical-semantic retrieval.

5.3. Qualitative Bucket Analysis

Table 7 classifies every top-1 retrieval result into the four quality buckets described in Section 4.3.

Table 7: Top-1 qualitative bucket analysis over all 377 test studies. *Clinical rescue* cases are the key validation of the false-negative mitigation objective.

Direction	Strict Hit	Clinical Rescue	Cluster Rescue	Miss
Image $\rightarrow$ Text	13 (3.4%)	64 (17.0%)	180 (47.7%)	120 (31.8%)
Text $\rightarrow$ Image	16 (4.2%)	93 (24.7%)	157 (41.6%)	111 (29.4%)
Combined	29 (3.8%)	157 (20.8%)	337 (44.7%)	231 (30.7%)

Of 754 total top-1 retrievals, **157 (20.8%) are clinical rescues**: the model retrieved the wrong patient but a clinically related study. An additional 337 (44.7%) are cluster rescues. Together, 523 / 754 top-1 results (69.4%) are at least clinically reasonable, with only 231 complete misses.

5.4. Training Dynamics

Key milestones observed in the validation curves:

- **Epoch 12:** clustering soft targets activated; cluster R@1 begins rising.
- **Epoch 15–25:** clinical weight ramps up; clinical-valid R@1 improves steadily.
- **Epoch 40:** cluster R@1 reaches its validation peak (65.25%).

- **Epoch 60:** clinical weight begins decay; strict R@1 starts recovering.
- **Epoch 90:** clinical-valid R@1 peaks on validation (52.11%).
- **Epoch 120: strict R@1 peaks on validation (4.91%)** — `best.pt` checkpoint saved.

The late strict peak at epoch 120, after the clinical weight has been decaying for 60 epochs, directly validates the curriculum design rationale.

6. Discussion

6.1. Interpretation of Clinical-Valid R@1

A clinical-valid mean R@1 of 59.86% over 147 queries per direction means that in more than half of cases where a clinically related result exists in the test set, the model retrieves it at rank 1. In a clinical workflow, retrieving a disease-matched case from a different patient is often more useful than retrieving the exact same study—a radiologist needs a visually and clinically similar reference, not a historical duplicate.

6.2. Why Strict R@1 Remains Low

Strict R@1 = 3.85% is modest, attributable to: (1) the small scale of IU-Xray (~3,000 training studies) relative to the task complexity; (2) high textual homogeneity—many normal reports contain nearly identical language, making exact discrimination difficult; and (3) the inherent trade-off between learning exact-pair alignment and disease-cluster semantics. We report this honestly as a known limitation rather than masking it with cluster or clinical-valid numbers alone.

6.3. Role of IDF Weighting and the Healthy Cluster

IDF weighting ensures that rare diseases (*e.g.* Pneumothorax 2.03%, Fracture 1.82%) contribute proportionally more to the similarity score than common diseases (*e.g.* Cardiomegaly 9.00%). This is clinically motivated: shared presence of a rare finding is a much stronger indicator of clinical relatedness than shared cardiomegaly. The healthy cluster bonus (Eq. 9) prevents the dominant normal class (57.88%) from fragmenting in embedding space, which would otherwise inflate cluster R@1 while hurting the discriminative representations.

6.4. Curriculum vs. Fixed Clinical Weight

The curriculum schedule (ramp → hold → decay) is critical. A fixed high w_c from training onset would pull the model toward coarse disease clusters before exact alignment has been established, degrading strict R@1 permanently. The epoch-120 strict peak occurring *after* 60 epochs of clinical decay empirically confirms that late-phase reduction of w_c allows the model to re-sharpen exact-pair discrimination.

6.5. Limitations

1. **Dataset scale.** IU-Xray contains only $\sim 3,000$ training studies—far smaller than large-scale medical CLIP datasets (BiomedCLIP: 15M pairs). Performance may improve substantially with more data.
2. **Weak pathology supervision.** Disease labels are generated by rule-based keyword matching, not physician annotation. Complex negation patterns may yield label noise.
3. **No published-baseline comparison.** V8 is compared only against an internal v7 baseline. Direct comparison with ConVIRT, GLORIA, or BiomedCLIP on the same split is left for future work.
4. **Single-dataset evaluation.** Generalization to larger datasets such as MIMIC-CXR has not been tested.
5. **Cluster metric inflation.** Cluster R@1 partially benefits from the large normal class; clinical-valid R@1 should be treated as the primary claim.
6. **Disabled components.** Prototype bank and hard negative mining are in the codebase but not activated in v8; ablation studies for these are deferred to future work.

7. Conclusion

We have presented a **clustering-guided contrastive learning framework** for chest X-ray image–report retrieval that directly addresses the false-negative problem inherent in standard contrastive training. The model combines a SwinV2-Base image encoder with Bio_ClinicalBERT, integrates multi-view fusion via attention pooling, and is trained with a three-phase curriculum that introduces and then decays clinical supervision.

On a fixed patient-stratified split of IU-Xray:

- **Clinical-valid mean R@1 = 59.86%** (primary metric, 147 queries/direction).
- **Cluster mean R@1 = 69.36%.**
- **Strict mean R@1 = 3.85%; MRR = 9.01%.**
- **157 clinical rescue cases** in top-1 (64 i2t + 93 t2i / 377 test studies).

The results validate the thesis that clustering-guided supervision substantially improves disease-level semantic retrieval while the curriculum preserves exact-pair alignment. Promising directions for future work include: systematic ablation of each proposed component (IDF Jaccard weighting, healthy cluster bonus, curriculum schedule), activation and evaluation of the prototype bank and hard negative mining modules, and cross-dataset generalization experiments on MIMIC-CXR.

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A. Full Hyperparameter Configuration (v8)

Table 8: Complete hyperparameter configuration of the primary v8 run.

Parameter	Value
<i>General</i>	
epochs / batch_size / grad_accum	150 / 32 / 4
effective batch size / seed	128 / 42
freeze_ep / eval_every	5 / 5
<i>Optimizer</i>	
lr_head / lr_enc / lr_text	2e-4 / 1e-5 / 1e-4
weight_decay / max_grad_norm	0.01 / 1.0
<i>Clustering</i>	
cluster_start / CLUSTER_ALPHA	12 / 0.35
proto_start (disabled)	999
rank_start / rank_weight (disabled)	999 / 0.0
mine_start / hn_frac (disabled)	999 / 0.0
<i>Clinical curriculum</i>	
clinical_start	15
ramp start weight / peak weight	0.03 / 0.12
ramp end / hold end	25 / 60
mid weight / decay end	0.08 / 105
final weight	0.04
<i>Loss weights</i>	
aux_weight	0.02

B. IDF Weights for Pathology Labels

Table 9: Relative IDF weights of the 12 pathology dimensions in the disease vector, derived from training-set prevalence (Table 3). Rare diseases receive higher weight, reflecting their greater discriminative clinical value.

Pathology	Prevalence (%)	Relative IDF weight
Pleural Other	0.89	Highest ■■■■■■
Fracture	1.82	Very high ■■■■■■
Pneumothorax	2.03	Very high ■■■■■■
Pneumonia	2.24	High ■■■■
Edema	2.53	High ■■■■
Support Devices	3.06	Medium ■■
Consolidation	5.49	Medium ■■
Pleural Effusion	5.67	Medium ■■
Cardiomegaly	9.00	Low ■
Atelectasis	10.00	Low ■
Lung Opacity	12.87	Low ■
Lung Lesion	13.85	Lowest □

C. Checkpoint Comparison on Test Set

Table 10: Comparison of all three saved checkpoints on the test set. `best.pt` is the primary checkpoint; `best_balanced.pt` coincides with it at epoch 120.

Checkpoint	Epoch	Strict Mean R@1	Cluster Mean R@1	Clin.-Valid Mean R@1
<code>best.pt</code>	120	3.85%	69.36%	59.86%
<code>best_balanced.pt</code>	120	3.85%	69.36%	59.86%
<code>best_clinical_valid.pt</code>	90	2.39%	65.78%	59.52%